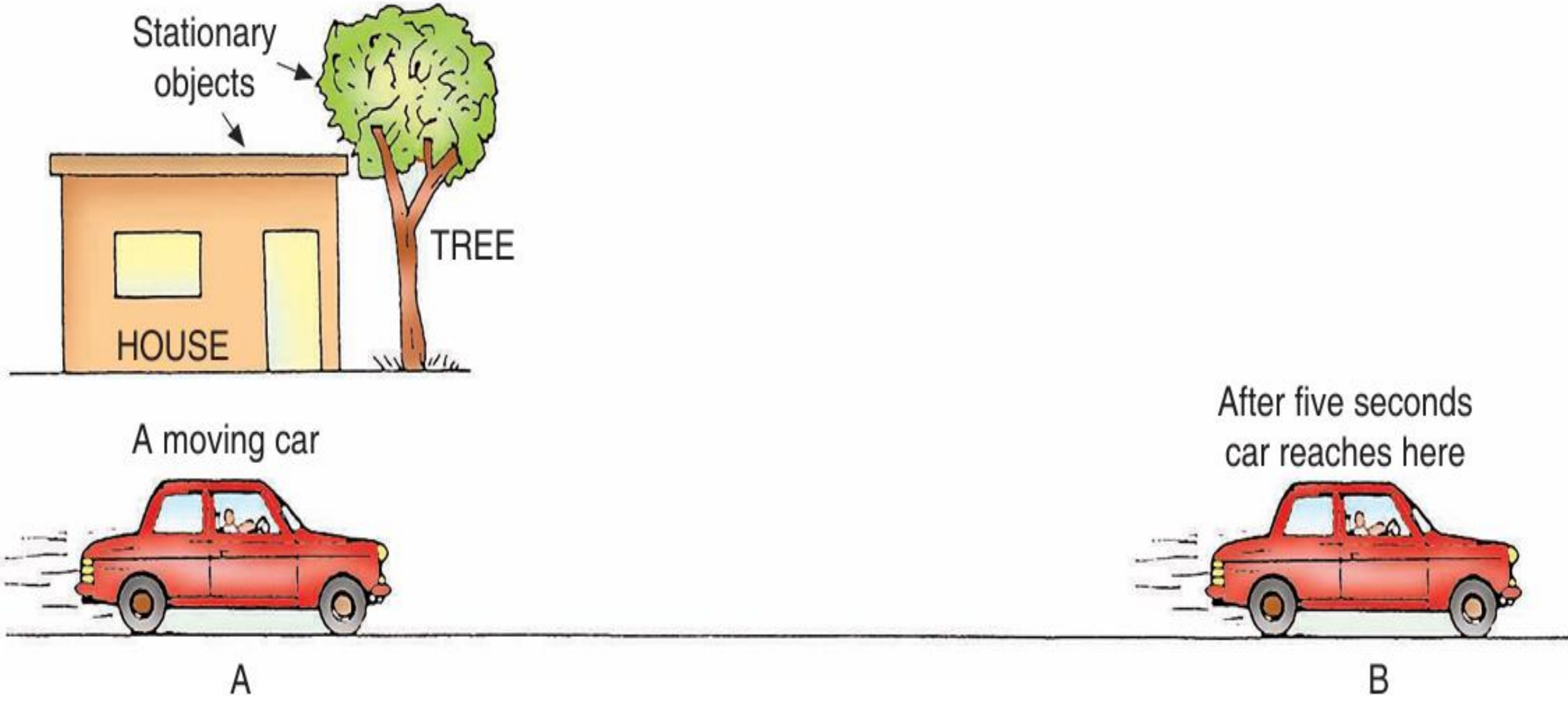
- We learned in our previous class about rest and motion.
- A body is said to be in a state of motion if its position, with respect to a fixed point or an object in its surroundings, keeps changing with time.
- A body is said to be at rest if it does not change its position with respect to a fixed point in its surroundings.

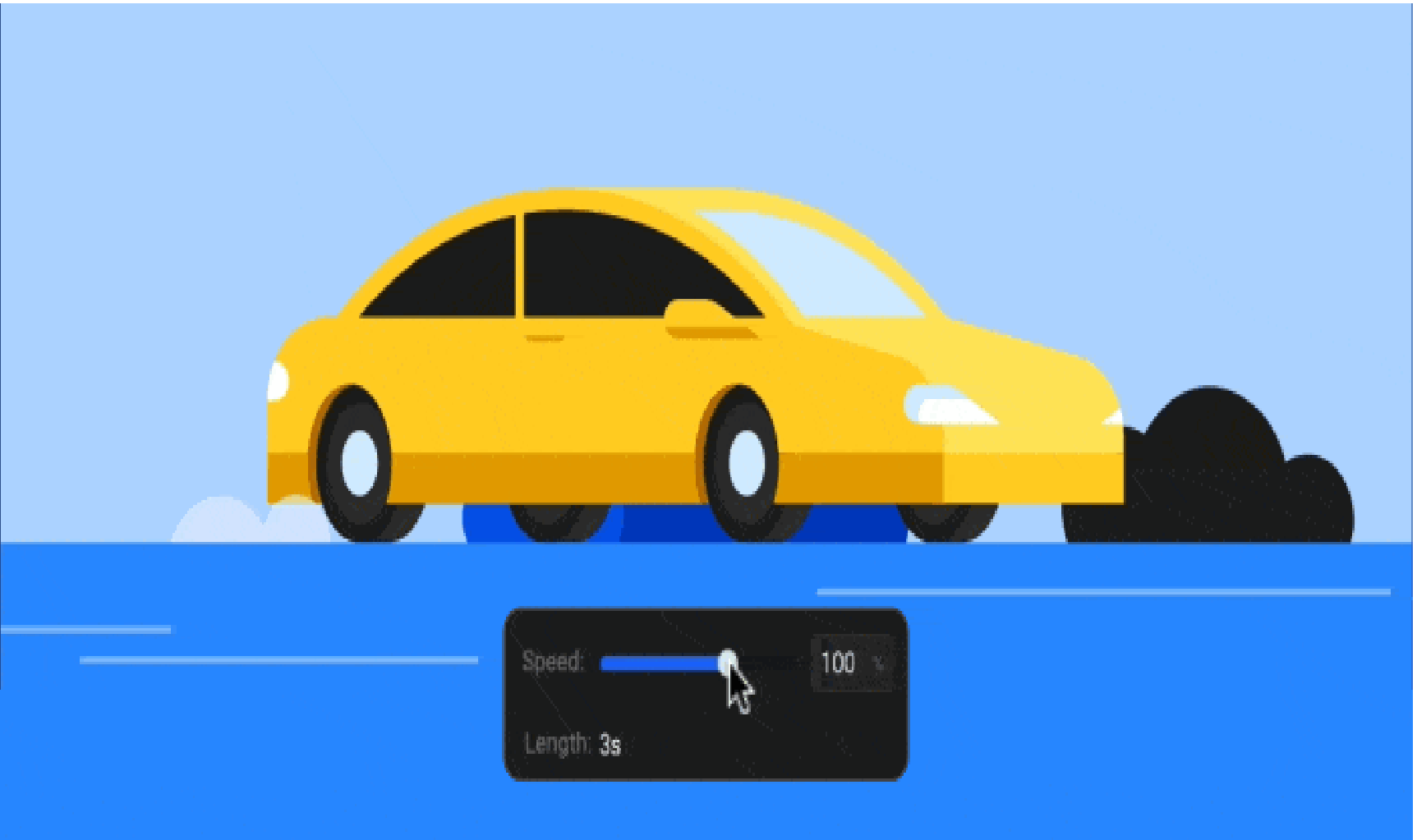

Stationary objects
TREE
HOUSE
A moving car
After five seconds car reaches here
A
B

FAST OR SLOW

- When we consider the motion of objects, we say that the motion of some objects is slower or faster than others.
- The distances moved by objects in a given interval of time help us to decide which object is moving faster or slower.
- For example, imagine that you and your friend are going to school.
- Suppose you start running at the same time as your friend begins to walk to the school.
- The distance covered by you after 10 minutes would be greater than that covered by your friend.
- Thus, the one who takes lesser time to cover the same distance is said to be moving faster than the other.
- Remember, that an object can be faster than one object and slower than some other object.
- We often say that the faster object has a higher speed.

THE CONCEPT OF SPEED

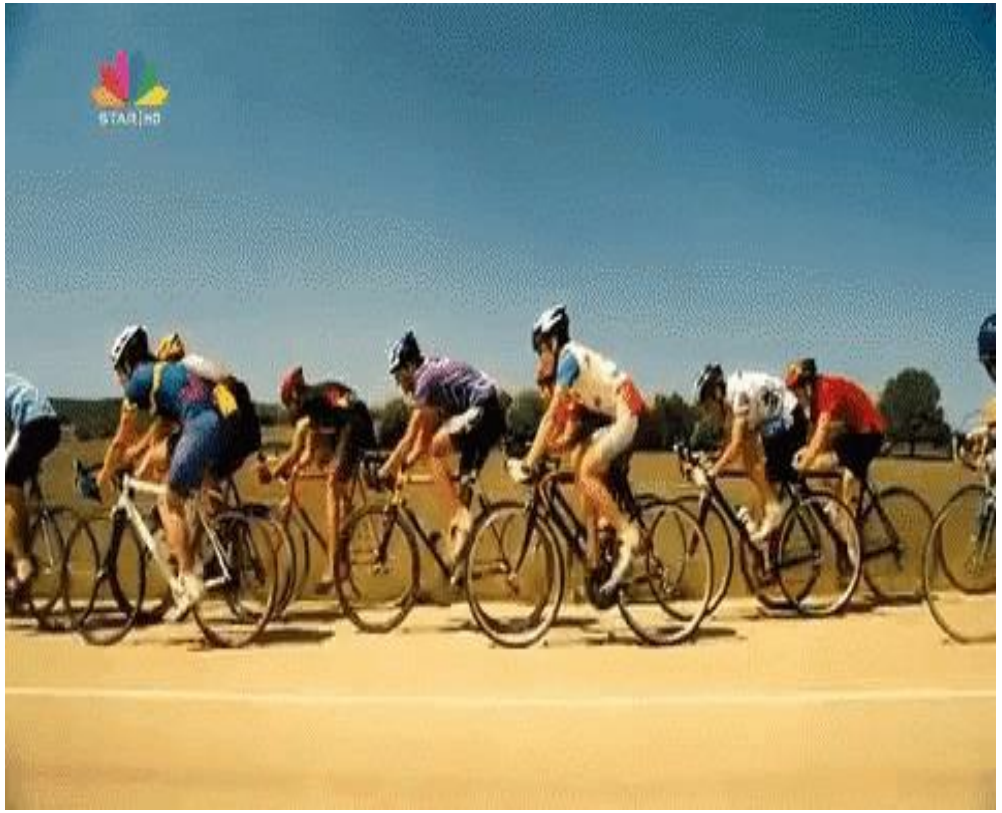
- The speed of a moving object can represent how fast or slow an object is moving.
- Speed is referred to as the distance covered by an object divided by the time taken to cover the distance.
- In other words, speed can be defined as the distance covered by an object per unit time.

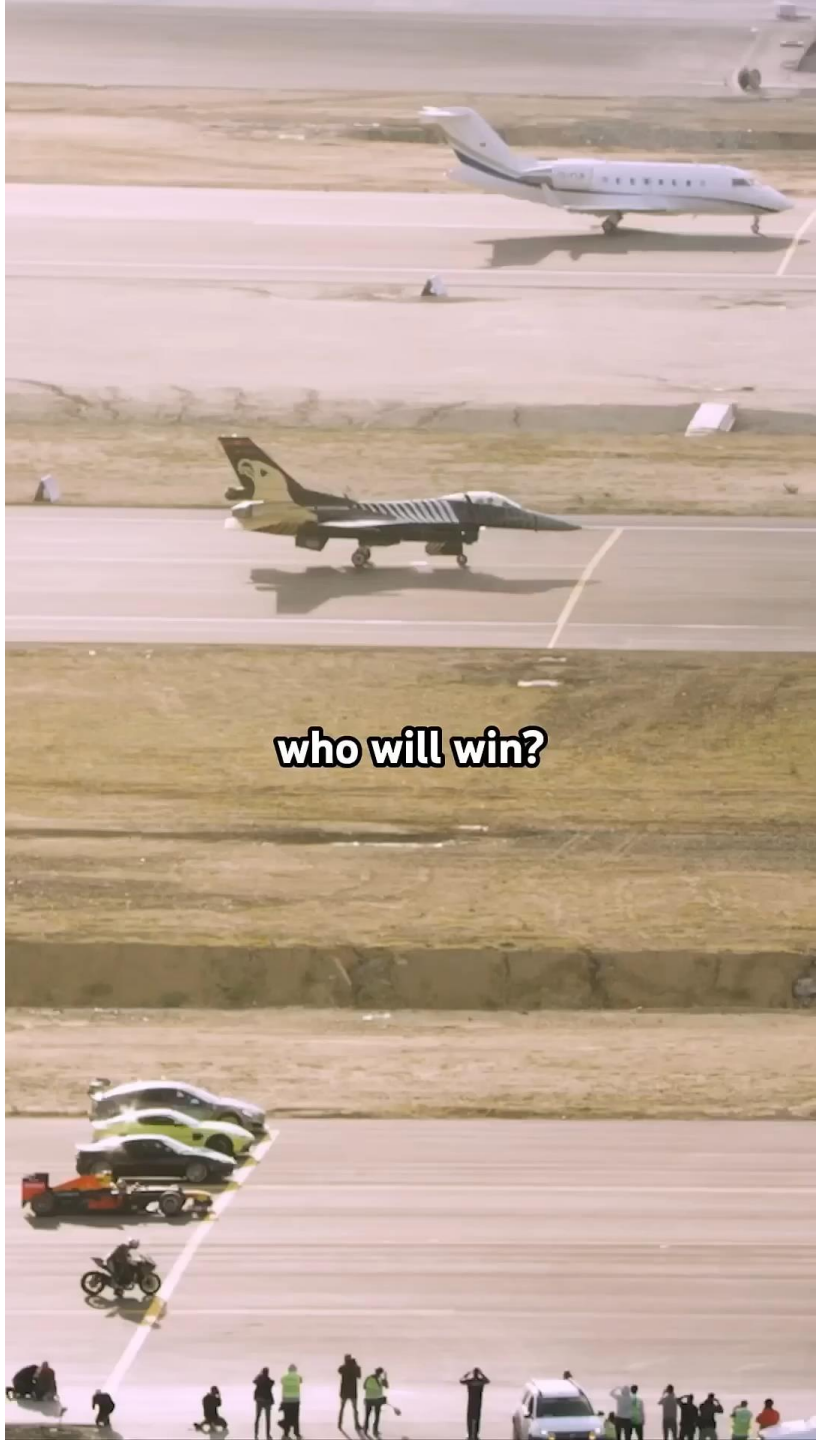


- Mathematically
- Speed = Distance travelled / Time taken

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

- The easiest way to determine which of the two given objects is moving faster is to compare the distances moved by them in a unit time, i.e., their speeds.
- Thus, if we know the distance covered by two objects in unit time, we can tell which one is faster.
- For example, if we say that a byecle is moving at a speed of 40 kilometres per hour, it means that the car will cover a distance of 40 kilometres in one hour.
- Similarly, another bicycle moving at a speed of 10 kilometres per hour indicates that it will cover a distance of only 10 kilometres in one hour.
- Clearly, the car is moving faster than the bicycle.





- A cheetah can have a speed of up to about 27 m/s (which is more than 97 km/h)

- A sprinter (fast runner) can have a speed of about 10 m/s (which is 36 km/h)

- A military jet aircraft can have a speed of about 500 m/s (which is 1800 km/h)

- Falcon – 320 km/h
- Blue Fish – 40–46 km/h
- Rabbit – 56 km/h
- Squirrel – 19 km/h
- Domestic mouse – 11km/h
- Human – 40 km/h
- Giant tortoise – 0.27 km/h
- Snail – 0.05 km/h

Average speed

- When we say that a car is moving at a speed of 40 kilometres per hour, it means that it will cover a distance of 40 kilometres in one hour.
- However, a car moving on a road seldom moves at a constant speed for one hour.
- In fact, it starts moving slowly and then picks up speed.
- Again during the travel it sometimes moves faster and sometimes slower

- So, when we say that a car has a speed of 40 kilometres per hour, we usually refer to the total distance covered by it in one hour.
- The speed calculated here is actually the average speed of the car.

- When we use the word speed, we generally refer to the average speed.
- Average speed of an object is the total distance covered divided by the total time taken to cover that distance.

$$\text{Average Speed} = \frac{\text{Total Distance Covered}}{\text{Total Time Taken}}$$

uniform motion & non uniform motion

UNIFORM MOTION AND NON-UNIFORM MOTION

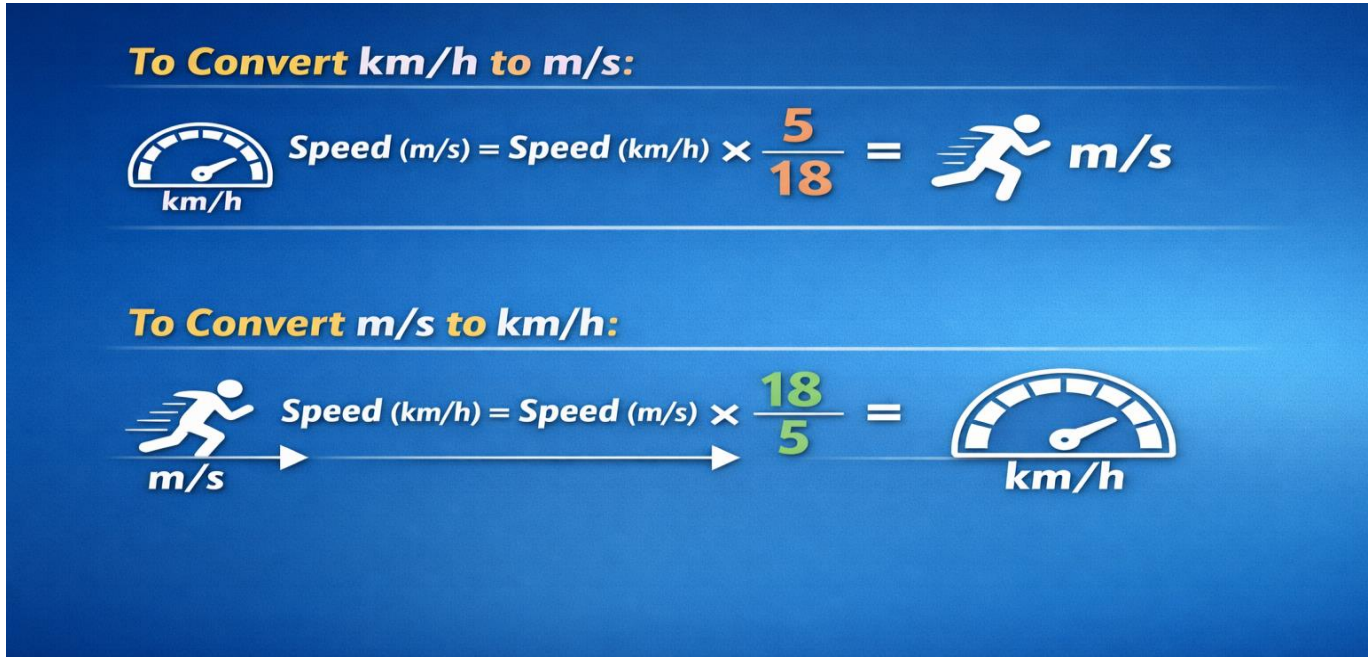
- When an object moves equal distances in equal intervals of time, we say that it is moving with a constant speed.
- In this case, the actual speed of the object at any given point of time is always the same as its average speed.
- This type of motion of an object with constant speed along a straight line path is called uniform motion.
- On the other hand, when an object moves unequal distances in equal intervals of time, we say that it is moving with a variable speed.
- In this case, the actual speed of the object at different points of time is different and deviates from the average speed as well.
- This type of motion of an object which keeps changing its speed along a straight line path is called non-uniform motion.

- A speedometer is a device fitted in a vehicle's dashboard that measures and displays the speed of a vehicle at a particular instant of time in km/h.
- The distance moved by a vehicle is measured by another device called an odometer.
- It records the distance travelled by the vehicle in kilometres.

Conversion of units of speed

- We need different scales of units to measure different speeds.
- For example, the speeds of rockets and satellites launched from the earth can be as large as 8 km/s, while the speed of a tortoise is barely about 8 cm/s.
- Let us understand how we can convert one unit of speed to another with the help of some examples.

(Recall that 1 hour has 60 minutes, and 1 minute has 60 seconds. Hence, 1 hour = 60 × 60 = 3600 seconds.)



- A cheetah can have a speed of up to about 27 m/s (which is more than 97 km/h)

Example : If a bus has travelled 480 km in 12 hours, find the speed of the bus.

Solution : We know that speed is defined as distance covered per unit time.

Speed = Distance / Time

= 480 km / 12 h

= 40 km/h

Convert

Example : Convert 54 km/h into m/s.

Solution: 54 km/h = (54 × 1000 m) / (60 × 60 s)

= 15 m/s

Example : Convert 20 m/s into km/h.

Solution:20 m/s = (20 × 60 × 60 km) / 1000 h

= 72 km/h

Example 4: Convert 108 km/h into m/s.

Solution:108 km/h = (108 × 1000 m) / (60 × 60 s)

= 30 m/s

1. **A body is said to be in motion when it changes its _____ with respect to a fixed point.**

   a) mass

   b) position

   c) shape

   d) size

b

2. **A body is said to be at rest when it does not change its _____ with respect to surroundings.**

   a) colour

   b) speed

   c) position

   d) temperature

c

3. **The distance moved by an object in unit time is called _____.**

   a) force

   b) speed

   c) work

   d) energy

b

4. Speed is mathematically given by _____.

a) Time ÷ Distance

b) Distance × Time

c) Distance ÷ Time

d) Distance – Time

c

5. If two objects cover different distances in the same time, the object covering more distance is _____.

a) slower

b) faster

c) at rest

d) uniform

b

6. When an object covers equal distances in equal intervals of time, the motion is called _____.

a) circular motion

b) non-uniform motion

c) uniform motion

d) oscillatory motion

c

7. When an object covers unequal distances in equal intervals of time, it shows _____.

a) uniform motion

b) non-uniform motion

c) periodic motion

d) rectilinear motion

b

8. The speed calculated using total distance and total time is called _____ speed.

a) instant

b) average

c) variable

d) constant

b

9. Average speed is equal to total distance divided by _____.

a) total mass

b) total time

c) total force

d) total velocity

b

10. A speedometer measures the _____ of a vehicle.

a) distance travelled

b) fuel level

c) instantaneous speed

d) engine power

c

11. An odometer measures the _____ travelled by a vehicle.

a) speed

b) distance

c) time

d) acceleration

b

12. If a car moves at 40 km/h, it means the car travels _____ in one hour.

a) 20 km

b) 30 km

c) 40 km

d) 80 km

c

13. A cheetah can run approximately at a speed of about _____.

a) 2 m/s

b) 10 m/s

c) 27 m/s

d) 100 m/s

c

14. A fast human sprinter can run at about _____.

a) 2 m/s

b) 5 m/s

c) 10 m/s

d) 50 m/s

c

15. A military jet aircraft can have a speed of about _____.

a) 50 m/s

b) 100 m/s

c) 500 m/s

d) 1000 m/s

c

16. To convert km/h into m/s, we multiply by _____.

a) 18/5

b) 5/18

c) 60

d) 3600

b

17. To convert m/s into km/h, we multiply by _____.

a) 5/18

b) 1/18

c) 18/5

d) 60

c

18. If speed changes during motion, the motion is called _____ motion.

a) uniform

b) circular

c) non-uniform

d) periodic

c